The Five Assumptions of Linear Regression

Contents

1. Overview 2
2. Assumption 1: a linear relationship 3
3. Assumption 2: no multicollinearity 4
    3.1 Why it is a problem 5
    3.2 Checking it 6
4. The residual plot 7
5. Assumption 3: normal residuals 8
6. Assumption 4: homoscedasticity 10
7. Assumption 5: no autocorrelation of the residuals 11
8. Summary 13
9. Sources 14
10. Key terms 14

Where this fits: Pipeline map step 8 (Model). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	**8 Model**	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Multicollinearity; Q-Q plot; Simple linear regression.

1. Overview

Key point: Linear regression is reliable when five conditions hold: a linear relationship, no **multicollinearity** (G-1273), normal residuals, constant spread of residuals, and no pattern in the residuals over the rows.

Linear regression always produces a line or hyperplane, whatever the data looks like. Whether that line can be trusted depends on a few conditions, called the **assumptions** (G-220) of linear regression. Like the rules of a game, they do not stop anyone from playing, but break them and the score means little.

Textbooks list them in slightly different ways; these five are the core ones (ISL §3.3.3; Kutner Ch. 3 and §7.6):

	Assumption	Checked with
1	Linear relationship between each feature and the output	scatter plots
2	No multicollinearity between the features	VIF, correlation heatmap
3	Residuals are normally distributed	histogram, Q-Q plot
4	Homoscedasticity: residuals have the same spread everywhere	residuals against predictions
5	No autocorrelation of the residuals	residuals in row order

Three words first. A **feature** (G-772) is an input variable (one column of the data table), an **observation** (G-1374) is one record (one row), and the **target** (G-1949) is the output we predict.

Recall from the equation of multiple linear regression the symbols used below. The model predicts:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots$$

Here $\hat{y}$ ("y hat") is the predicted target, y the actual target, $x_1, x_2, \dots$ the features and each β a coefficient (a learned number). The first two assumptions concern the features; the last

three concern the **residuals** (G-705), the errors $y_i - \hat{y}_i$ on each observation. Figure 1 draws them for the model below (each dot is one observation: its actual target against the model's prediction): each stick is the gap between an observation's actual target and the model's prediction.

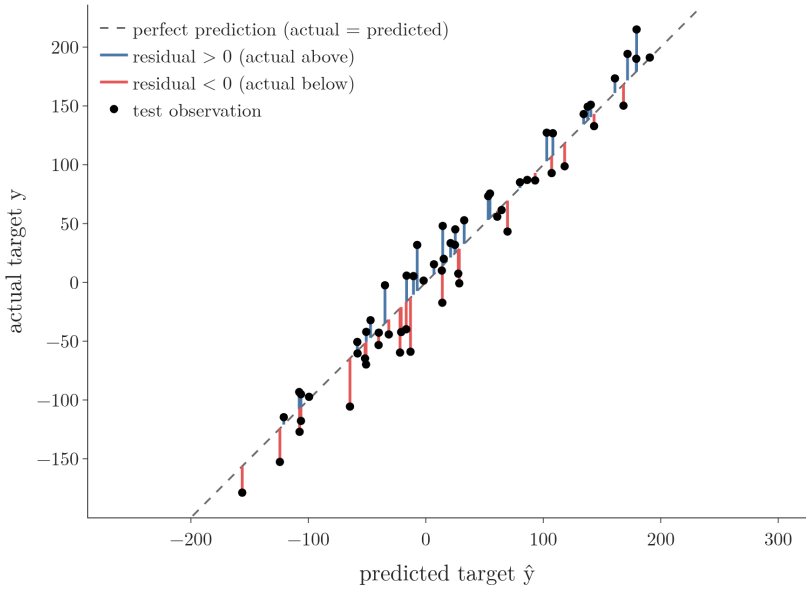


Figure 1: The 60 test observations: actual target against predicted target. Each stick is one residual, the vertical gap to the dashed diagonal of perfect predictions (blue: actual above, red: actual below). Assumptions 3 to 5 are about these sticks.

The example data has 200 observations, three features and one target. A linear regression trained on 70% of it scores $R^2 = 0.96$ on the other 30% (60 observations); each check below uses those 60 test residuals.

Python: The model whose assumptions we check.

```
df = pd.read_csv("data/data.csv")
X, y = df.iloc[:, :3].to_numpy(), df["target"].to_numpy()
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1)
model = LinearRegression().fit(X_train, y_train)
residual = y_test - model.predict(X_test)
```

2. Assumption 1: a linear relationship

Key point: Each **feature** (G-772) should relate to the output along a straight line, rising or falling. A curved relationship needs a different model.

Linear regression fits straight lines, so the relationship between each feature and the output should be roughly linear. Both directions are fine:

- **positive:** the output rises as the feature rises;
- **negative:** the output falls as the feature rises.

A curved relationship (for example, the output growing with the square of the feature) breaks this **assumption** (G-220), because a straight line then misses the pattern systematically.

The check is a scatter plot of the output against each feature (Figure 2).

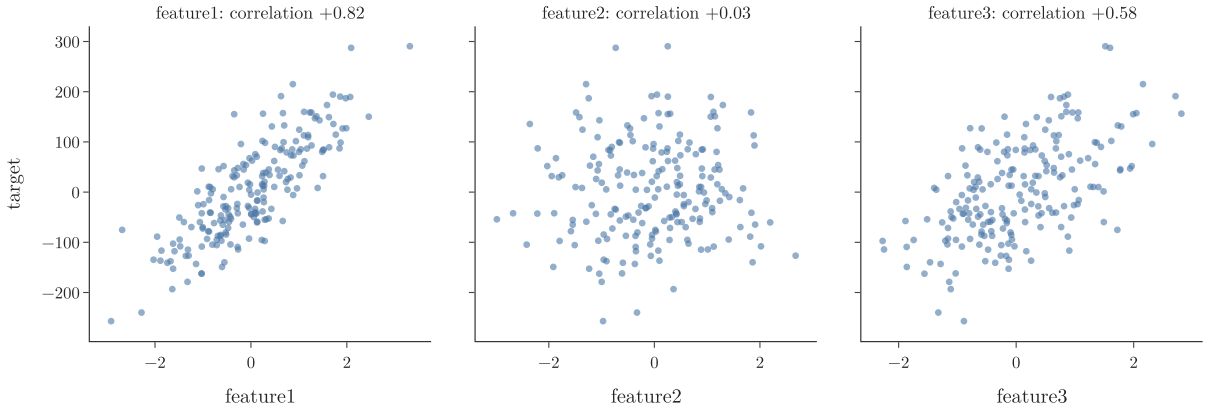


Figure 2: Each feature against the target

- **feature1:** a clear rising line (**correlation** (G-490) 0.82).
- **feature3:** also rising, with more scatter (0.58).
- **feature2:** no visible relationship at all (0.03). Its fitted coefficient is only -0.28, against 72.7 and 53.3 for the other two: the model has learned to almost ignore it.

> **Extra:** A feature with no relationship, like feature2, does not break the assumption so much as add nothing; the model gives it a near-zero weight. When a feature has a curved relationship, a simple approach is to add non-linear transformations of that feature, such as $\log x$, $\sqrt{x}$ or x^2, to the model (ISL §3.3.3). Adding x^2 is **polynomial regression** (G-1515) (adding powers as new features).

3. Assumption 2: no multicollinearity

> **Key point:** The **features** (G-772) should not be strongly related to each other. If they are, the model cannot tell which one deserves the credit, and the coefficients become unreliable.

Multicollinearity (G-1273) means one feature can be (largely) predicted from the others. Multicollinearity was met as the dummy variable trap in one-hot encoding.

3.1 Why it is a problem

> **Key point:** Each coefficient measures the effect of its **feature** (G-772) with the others held fixed. Related features cannot be held fixed separately.

A coefficient β_1 is the change in the output when x_1 rises by 1 *and all other features stay the same*. If x_1 and x_3 always move together, "change x_1 and keep x_3 fixed" never happens in the data. The model then cannot separate their effects.

An analogy: two scientists finish a project together. If one is a physicist and the other a chemist, it is easy to say who contributed what. If both have identical skills, it is impossible. With related features the model faces the second case, so the coefficients become unreliable: a small change in the data can move them a long way (ISL §3.3.3). The predictions, on the other hand, usually stay good (Kutner §7.6).

Figure 3 tests this on our data. We refit the model on 20 **bootstrap** (G-320) resamples of the training data: each resample draws 140 **observations** (G-1374) from the 140 with replacement, so it differs a little from the original. With the Note's own features (left), feature1's coefficient stays between 69.9 and 78.0. Then we add a near-copy of feature1, feature1 plus a little noise (VIF 429, Section 3.2). Now feature1's coefficient swings from 21.5 to 102.5 and the copy's swings the opposite way, sometimes below 0. Their sum stays between 69.7 and 78.0: the model knows the pair's joint effect, but not how to split the credit.

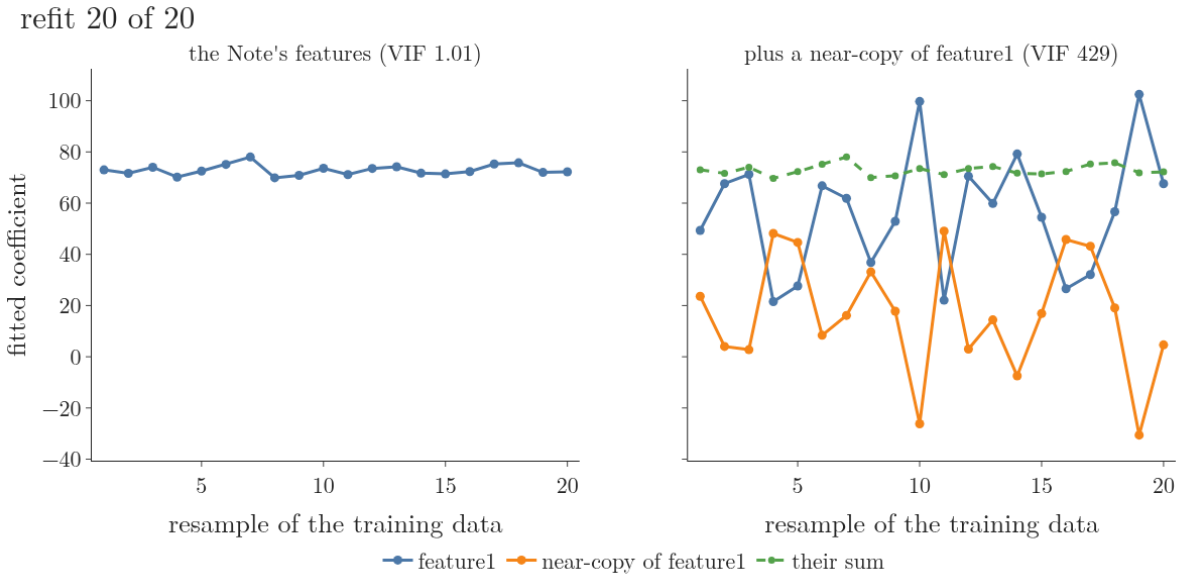


Figure 3: Refitting on 20 bootstrap resamples of the training data. Left: the Note's features; feature1's coefficient barely moves. Right: with a near-copy of feature1 added, the two coefficients swing wildly in opposite directions, while their sum (dashed green) stays steady.

3.2 Checking it

> **Key point:** A VIF near 1 means no **multicollinearity** (G-1273); above 5 is a problem. All three features here have VIF 1.01.

The standard check is the **variance inflation factor (VIF)** (G-2075). For each **feature** (G-772), a regression predicts that feature from all the other features; if they predict it well (high R^2), the feature is redundant.

In words: the VIF of a feature is one divided by the part of it the other features cannot explain.

$$\text{VIF of feature } j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R^2 of predicting feature j from the others. With numbers: if the others explain 80% of a feature ($R_j^2 = 0.8$), its VIF is the usual danger line:

$$\frac{1}{1 - 0.8} = \frac{1}{0.2} = 5$$

If they explain none of it, the VIF is 1.

> **Python:** VIF with statsmodels.

```python
from statsmodels.stats.outliers_influence import (
    variance_inflation_factor)
from statsmodels.tools import add_constant

Xc = add_constant(X_train)      # add an intercept column
[variance_inflation_factor(Xc, i) for i in range(1, 4)]
# [1.011, 1.010, 1.014]
```

A quicker, rougher check is a **heatmap** (G-886) of the **correlations** (G-490) between the features (Figure 4, left): a table of correlations in which each cell pairs the feature of its row with the feature of its column, coloured darker the closer the value is to 1. The diagonal is 1 because each feature matches itself. Here every correlation between different features is at most 0.06: no multicollinearity.

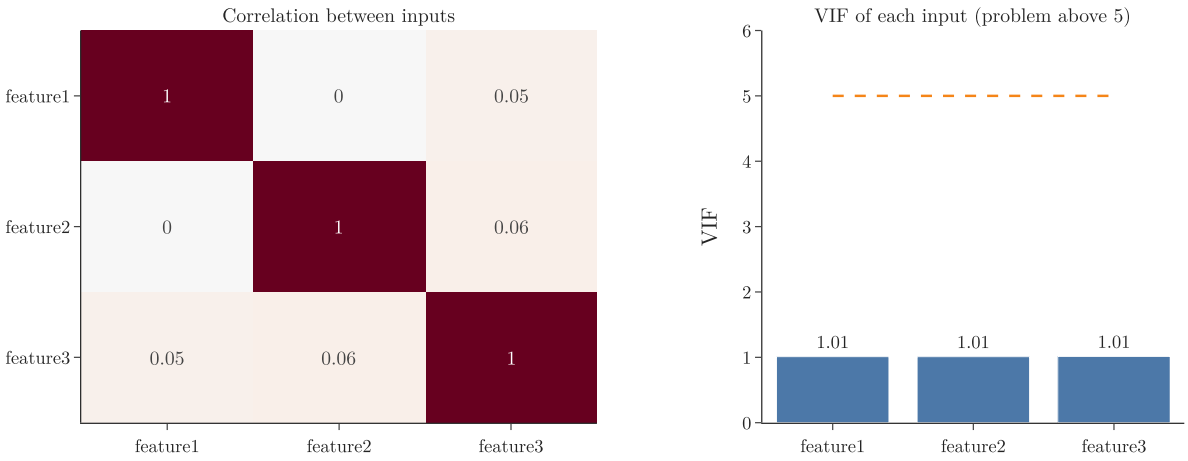


Figure 4: Correlations between the features, and their VIF

> **Extra:** If VIF is high, two simple fixes are to drop one of the related features, or to combine them into one feature, such as their average after scaling (ISL §3.3.3). Ridge regression (the idea: penalise large coefficients) was also designed for related features (Hoerl and Kennard).

4. The residual plot

> **Key point:** A residual plot puts every **residual** (G-705) above or below a flat zero line. Random scatter around zero means the model fits; any shape means an assumption is broken.

The last three assumptions are all checked on one kind of picture, so we build it by hand first. Take four points, $(1, 2)$, $(2, 3)$, $(3, 7)$ and $(4, 8)$. Their least-squares line is $\hat{y} = 2.2x - 0.5$.

1. **Compute each residual,** actual minus predicted:

x	actual y	predicted $\hat{y} = 2.2x - 0.5$	residual
1	2	1.7	$+0.3$
2	3	3.9	-0.9
3	7	6.1	$+0.9$
4	8	8.3	-0.3

2. **Draw a flat line at 0.** It stands for the fitted line.
3. **Plot each residual** at its x: above the zero line when the point lies above the fitted line, below when it lies below.

Figure 5 does these steps one point at a time. The result is the **residual plot** (G-2207): the fitted line laid flat, with only the errors left to look at.

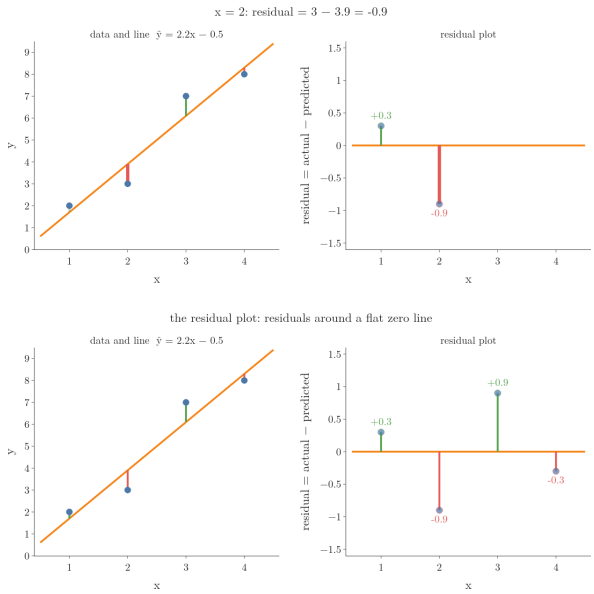


Figure 5: Building a residual plot from four points. Left: the points, the least-squares line and each residual as a stick (green above the line, red below). Right: the same sticks moved onto a flat zero line. Idea after Khan Academy, "Residual plots".

How to read a residual plot:

- **Points scattered evenly above and below 0, with no shape:** the line describes the data well.
- **A curve,** for example the residuals sit high on the left, low in the middle and high again on the right (above the dashed zero line, then below it, then above again): the relationship is not a straight line. Assumption 1 is broken, and a non-linear model such as **polynomial regression** (G-1515) fits better (Figure 7, left).
- **A funnel or a wave:** assumptions 4 and 5 below.

With several **features** (G-772) there is no single x to put on the horizontal axis, so the predicted value $\hat{y}$ is used instead. The reading rules stay the same.

5. Assumption 3: normal residuals

> **Key point:** The **residuals** (G-705) should follow a bell-shaped (normal) distribution centred on 0: most errors small, few large, positive and negative alike.

When the model is right on average, its errors should scatter around 0: many small errors and few large ones, as often too high as too low. Such a bell-shaped spread is a **normal distribution** (G-1343).

Why it matters: the ranges and tests reported with the model rest on this **assumption** (G-220). A **prediction interval** (the range a new **observation** (G-1374)'s target is expected to fall in) depends strongly on normal residuals, so with clearly non-normal residuals it

becomes inaccurate. The **confidence intervals** (G-446) (ranges likely to hold the true coefficient) and tests for the coefficients are more forgiving, especially with a large sample (Penn State STAT 501, Lessons 4.1, 7.1 and 7.2).

Non-normal residuals do not make the coefficients themselves wrong on average. Least squares gives unbiased coefficients as long as the errors average zero, whatever their shape; the Gauss–Markov theorem says it is even the best linear unbiased estimator without normal errors (Wikipedia, "Gauss–Markov theorem"). So normality matters for the intervals and tests, not for the fitted line.

Two checks, both on the residuals (Figure 6, top):

- **Histogram or density plot:** roughly a bell centred on 0. Here it is, apart from a small bump.
- **Q-Q plot** (G-1596; see how a Q-Q plot is built): each residual is plotted against the value a perfect bell would give, so the points should lie on the straight line. Here they do, with small wiggles.

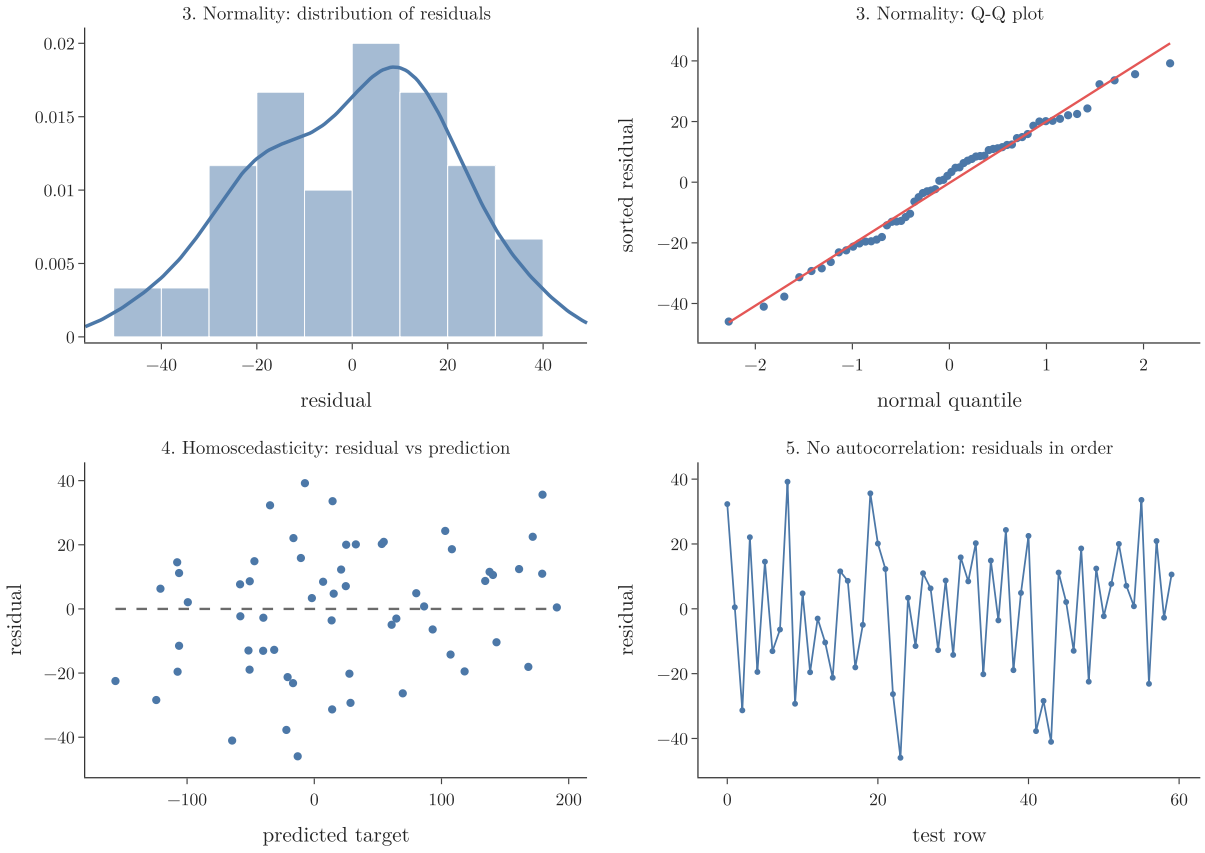


Figure 6: Residual checks for assumptions 3, 4 and 5

Extra: Formal tests exist as well. The **Shapiro-Wilk test** (G-1788) gives a **p-value** (G-1433; the chance of results at least this extreme if the data were normal, see the definition); above 0.05 means no evidence against normality. Here $p = 0.51$, and the skewness of the residuals is -0.23, close to 0. With large samples these tests flag even small departures from normality, so look at the plots too (Ghasemi and Zahediasl 2012).

6. Assumption 4: homoscedasticity

Key point: The residuals should have the same spread for small and large predictions. A funnel shape (heteroscedasticity) breaks the **assumption** (G-220).

Homoscedasticity (G-902) means "same scatter": the size of the errors does not depend on the size of the prediction. Its opposite, **heteroscedasticity** (G-889), is common (ISL §3.3.3): for example, a house-price model that is off by a few thousand on cheap houses and by much more on expensive ones.

The check is a scatter plot of **residuals** (G-705) against predicted values (Figure 6, bottom left). The plot should look like an even band around 0, with no shape. Here it does: the spread is roughly the same from the lowest to the highest prediction.

Figure 7 (middle) shows what heteroscedasticity looks like: a funnel that widens to the right.

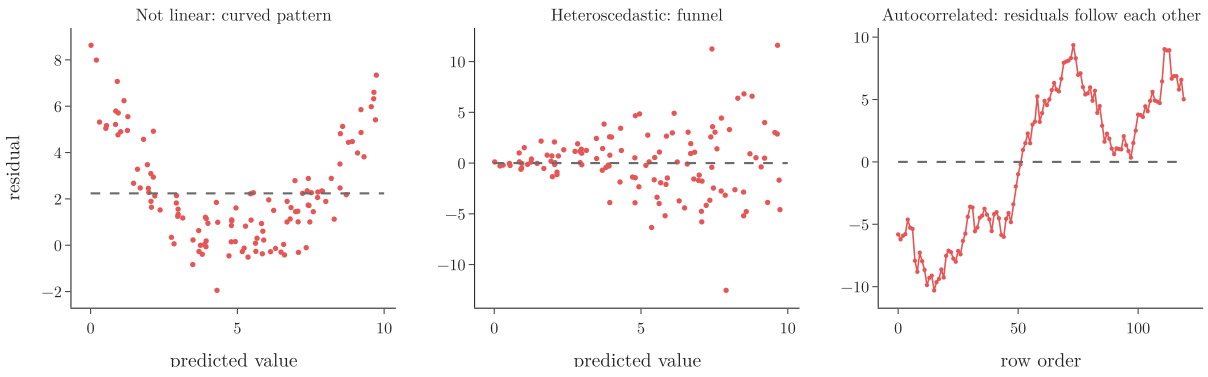


Figure 7: What failed assumptions look like (made-up data)

Extra: Heteroscedasticity does not make the coefficients wrong on average, but it makes the model's uncertainty estimates wrong: the confidence intervals and p-values that statistics packages report (Wooldridge §8.1). One fix is to take the log or square root of the **target** (G-1949). The log shrinks large values more than small ones, and can turn a funnel into an even band (ISL §3.3.3).

7. Assumption 5: no autocorrelation of the residuals

> **Key point:** One **residual** (G-705) should not predict the next. Plotted in row order, the residuals should jump around randomly, not drift in waves.

Autocorrelation (G-230) means each residual is related to the one before it: if the model was too high on one **observation** (G-1374), it is also too high on the next. Autocorrelation is common in **time series** (G-1975) data, measurements taken at points in time (ISL §3.3.3).

The check is to plot the residuals in row order (Figure 6, bottom right). They should jump up and down with no pattern. Here they do.

Figure 7 (right) shows positive autocorrelation: long runs above 0 followed by long runs below, like a slow wave. One cause is a missing feature that changes slowly from one observation to the next, such as a time trend: the model cannot see the trend, so the trend shows up in the residuals. Figure 8 shows this test (details in the Extra below): residuals of a model that misses a slow wave form the same wave; adding the wave as a feature leaves random jumps. Autocorrelation makes the model look more certain than it is: its **confidence intervals** (G-446) come out too narrow (ISL §3.3.3).

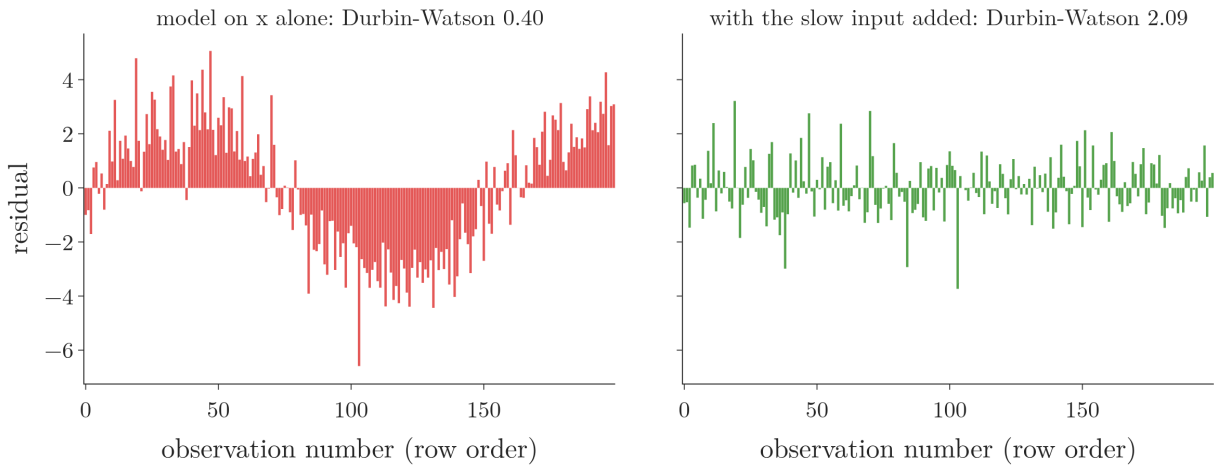


Figure 8: Residuals in row order for data built as y = 2x + 3 sin(t/25) + noise. Left: a model on x alone leaves the slow wave in its residuals (Durbin-Watson 0.40). Right: with sin(t/25) added as a feature, the residuals jump randomly (Durbin-Watson 2.09).

> **Extra:** The **Durbin-Watson statistic** (G-649) puts a number on autocorrelation: about 2 means no autocorrelation, towards 0 means positive autocorrelation, towards 4 negative (statsmodels docs). Here it is 2.31, close to 2: no sign of autocorrelation. To test the missing-trend cause, the notebook makes data from this rule, with t the observation number:
>
> $$y = 2x + 3\sin(t/25) + \text{noise}$$

Fitting on x alone gives Durbin-Watson 0.40; adding the slow wave $\sin(t/25)$ as a feature brings it back to 2.09. Figure 7 (left) also shows a residual plot when **assumption** (G-220) 1 fails: a curve instead of a flat band.

8. Summary

Assumption	What it means	How to check	This data	Why it matters
1. Linearity	each feature relates to the output along a line	scatter plots	yes (feature2 unrelated)	a straight line misses a curved pattern every time
2. No multicollinearity	features not related to each other	VIF (problem above 5), correlation heatmap	VIF 1.01	related features make the coefficients swing (21.5 to 102.5 with a near-copy)
3. Normal residuals	errors form a bell around 0	histogram, Q-Q plot	yes	a model right on average makes many small errors, as often too high as too low; prediction intervals rely on it, and so do coefficient tests in small samples
4. Homoscedasticity	errors have the same spread everywhere	residuals vs predictions	yes	a funnel makes the reported confidence intervals and p-values wrong
5. No autocorrelation	errors do not follow each other	residuals in row order	yes	a wave means a missing feature, and the model looks more certain than it is

- **Assumptions** (G-220) 1 and 2 are about the features; 3 to 5 are about the residuals, so the first two are checked on the data and the last three after fitting.
- A failed assumption does not stop the model from running; it makes its coefficients or uncertainty estimates untrustworthy.
- **Residual plots** (G-2207) should look like random noise; any shape (curve, funnel, wave) points to a broken assumption, because a good fit leaves only random errors behind.
- Together these answer the opening question: the line can be trusted when these five conditions hold, and each one has its own plot to check it.

9. Sources

Built from

- CampusX, "What are the main Assumptions of Linear Regression? | Top 5 Assumptions of Linear Regression", YouTube, https://www.youtube.com/watch?v=EmSNAtcHLm8
- Khan Academy, "Residual plots", YouTube, https://www.youtube.com/watch?v=VamMrPZ-8fc (building a residual plot by hand and reading it, section 4)

Other references

- **ISL**: James, Witten, Hastie, Tibshirani, *An Introduction to Statistical Learning*, 2nd ed., Springer, 2021. §3.3.3, Potential Problems.
- **Kutner**: Kutner, Nachtsheim, Neter, Li, *Applied Linear Statistical Models*, 5th ed., McGraw-Hill, 2005. Ch. 3 (diagnostics and remedial measures), §7.6 (multicollinearity and its effects).
- **Hoerl and Kennard**: A. E. Hoerl and R. W. Kennard, "Ridge Regression: Biased Estimation for Nonorthogonal Problems", *Technometrics* 12(1), 55–67, 1970.
- **Ghasemi and Zahediasl**: A. Ghasemi and S. Zahediasl, "Normality Tests for Statistical Analysis: A Guide for Non-Statisticians", *International Journal of Endocrinology and Metabolism* 10(2), 486–489, 2012.
- **Wooldridge**: J. M. Wooldridge, *Introductory Econometrics: A Modern Approach*, Cengage. Chapter 8, Heteroskedasticity (§8.1, consequences for OLS).
- **Penn State STAT 501**: Penn State Eberly College of Science, *STAT 501: Regression Methods*, online notes, online.stat.psu.edu/stat501. Lesson 4.1 "Background" (tests and confidence intervals for the coefficients are fairly robust to departures from normality; prediction intervals are quite sensitive), Lesson 7.1 "Confidence Interval for the Mean Response" (with a large sample the errors can deviate substantially from normality) and Lesson 7.2 "Prediction Interval for a New Response" (depends strongly on normal errors).
- **Gauss–Markov**: Wikipedia, "Gauss–Markov theorem", en.wikipedia.org/wiki/Gauss–Markov_theore (least squares is the best linear unbiased estimator when the errors have mean zero, equal variance and are uncorrelated; the errors do not need to be normal).
- **statsmodels docs**: statsmodels documentation, `statsmodels.stats.stattools.durbin_watson`.

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Assumption (of a model) (G-220)	A condition the data must meet for the model's results to be reliable.

Term	Meaning
Variance inflation factor (VIF) (G-2075)	A number for each input column that shows how well the other inputs predict it, $1/(1 - R_j^2)$; above 5 signals multicollinearity (the input is redundant).
Residual plot (G-2207)	The residuals (errors) plotted above and below a flat zero line, against a feature or the predicted value; a funnel or a wave in it shows that an assumption of linear regression is broken.
Homoscedasticity (G-902)	Prediction errors (residuals) with the same spread for every predicted value; an assumption of linear regression, checked by plotting residuals against predictions.
Heteroscedasticity (G-889)	Prediction errors (residuals) whose spread changes with the predicted value, often widening like a funnel; it breaks the equal-spread assumption (homoscedasticity) of linear regression.
Autocorrelation (G-230)	When each residual is related to the one before it in row order (too high on one observation, too high on the next), as is common in time series; linear regression assumes there is none.
Durbin-Watson statistic (G-649)	A number from 0 to 4 that checks whether a regression's residuals are linked to the residuals next to them (autocorrelation); about 2 means no such link.
Linear relationship (G-1095)	A relationship between two columns that follows a straight line.
statsmodels (G-1885)	A Python library for statistical models and tests, such as the VIF and the Durbin-Watson statistic used to check the assumptions of linear regression.
Multicollinearity (G-1273)	A relationship between input columns, so that one can be calculated (or largely predicted) from the others; it makes a linear model's coefficients unstable and hard to interpret.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.

Term	Meaning
Target (G-1949)	The output column we predict, such as the class; also called the label.
Error (residual) (G-705)	The gap between an actual value and the model's prediction on one data point: actual minus predicted.
Correlation (G-490)	A number from -1 to +1 that says how strongly two numerical features rise or fall together along a straight line: the covariance divided by both standard deviations, so the units drop out.
Polynomial regression (G-1515)	Linear regression on powers (and products) of the inputs, to fit curves.
bootstrap (G-320)	BaggingClassifier setting: draw rows with replacement (True, bagging) or without (False, pasting).
Heatmap (G-886)	A table drawn as coloured cells, darker for larger values.
Normal distribution (G-1343)	A symmetric, bell-shaped distribution: most values sit near the mean and they get rarer the further out we go; its mean and standard deviation set its centre and width.
Confidence interval (G-446)	A range of values, computed from a sample, that we expect to contain a population parameter such as the mean, so it shows how uncertain the estimate is. The method captures the true value in a stated share of repeated samples, such as 95%.
Q-Q plot (G-1596)	A plot of a column's sorted values against the values a theoretical distribution, often the normal, would have; points on the line mean the data follows it.
Shapiro-Wilk test (G-1788)	A statistical test of whether data follows a normal distribution: it takes the values and returns a statistic and a p-value, and a small p-value says the data is unlikely to be normal.
P-value (G-1433)	The probability, assuming H_0 is true, of getting a sample as or more extreme than ours.
Time series (G-1975)	Data recorded at regular time intervals.